SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING COMPUTER SCIENCE AND ENGINEERING LIST OF EXPERIMENTS

SUB CODE & NAME: CSA1712-ARTIFICIAL INTELLIGENCE

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1.Write the python program to solve 8-Puzzle problem

Code:

```
from collections import deque

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

queue = deque([(start, [])])
visited = {start}

while queue:
    state, path = queue.popleft()

    if state == goal:
        print("Moves:", path)
        break

    p = state.index(0)
    moves = []

    if p >= 3:
        moves.append(p - 3)
    if p < 6:
        moves.append(p + 3)
    if p % 3 != 0:
        moves.append(p - 1)
    if p % 3 != 2:
        moves.append(p + 1)

    for x in moves:
        new = list(state)
        new[p], new[x] = new[x], new[p]
        new = tuple(new)

        if new not in visited:
            visited.add(new)
            queue.append((new, path + [new]))
```

Output:

```
== RESTART: C:/Users/Lekha/AppData/Local/Programs/Python/Python313/ai 1
Moves: [(1, 2, 3, 4, 5, 6, 7, 0, 8), (1, 2, 3, 4, 5, 6, 7, 8, 0)]
```